frontiers | Frontiers in Virtual Reality

TYPE Original Research
PUBLISHED 12 February 2025
DOI 10.3389/frvir.2025.1515937

Check for updates

OPEN ACCESS

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RECEIVED 23 October 2024
ACCEPTED 24 January 2025
PUBLISHED 12 February 2025

CITATION
Micheletto V, Accardi S, Fici A, Piccoli F, Rossi C, Bilucaglia M, Russo V and Zito M (2025) Enjoy it! Cosmetic try-on apps and augmented reality, the impact of enjoyment, informativeness and ease of use.
Front. Virtual Real. 6:1515937.
doi: 10.3389/frvir.2025.1515937

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Enjoy it! Cosmetic try-on apps and augmented reality, the impact of enjoyment, informativeness and ease of use

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Virtual Try-On cosmetics apps based on Augmented Reality (AR) technology can improve both consumer product evaluation and purchase decisions, while also supporting companies' marketing strategies. This study explores the factors influencing the use of AR-based cosmetics apps by administering the Technology Acceptance Model (TAM) and additional scales on a sample of 634 Italian consumers. Perceived Informativeness (PI) and Perceived Ease of Use (PEOU) were hypothesized as predictors of TRUST, DOUBT, Makeup Involvement (MI), Perceived Diagnosticity (PD), and Behavioral Intention (BI), with Perceived Enjoyment (PE) acting as a mediating variable. The structural equation model (SEM) confirmed PI as a strong predictor, with PE serving as a key mediator. The findings suggest that a moderate level of PE and PEOU is ideal - excessive simplicity or playfulness increases DOUBT and decreases TRUST. Both PD and BI are positively affected by the AR experience, with their coexistence being crucial for effective app usage. Additionally, PI, mediated by PE, significantly influences BI, emphasizing the role of information in consumer decision-making. These results provide valuable insights for the cosmetics industry, offering guidance to refine user experiences and enhance consumer engagement and satisfaction.

KEYWORDS

augmented reality, cosmetics, beauty, trust, perceived enjoyment, virtual try-on apps, technology acceptance model, percieved informativeness

1 Introduction

Augmented Reality (AR) technology combines real images/environments, captured by the camera of PC, mobile phones or other smart devices, with virtual 3D digital graphics/objects, making them coexist in the same space at the same time (Azuma, 1997; Fox, 2012; Kounavis et al., 2012; Rese et al., 2017; Russo et al., 2022). It can be implemented on cutting-edge devices such as Apple Vision Pro or Spacetop AR laptop, as well as on mobile apps for general purpose off-the-shelf smartphones (Arena et al., 2022). Reduced costs and technological advances are expanding AR globally: worldwide revenues are expected to surpass US$198 billion in 2025, and the number of users to exceed US$1.33 billion (Rejeb et al., 2023). According to research data from the IDC Worldwide Augmented and Virtual Reality Spending Guide, the consumer sector accounted for 53% of the market share of AR-related spending in 2020 (Hsu et al., 2021).

and Wilson, 2019; tom Dieck et al., 2023), and behavioural intention to use AR (Oyman et al., 2022; Rese et al., 2017) and not considering the complex interconnection of other psychological variables that may arise.

To address these limitations, we proposed a model based on different predictors and mediators, as well as multiple outcomes, considering also the potential psychological and behavioral barriers that might lead consumers to distrust e-commerce platforms employing AR, abandoning the app and/or opting to visit physical stores to finalize their purchases after identifying the desired product (Kim and Han, 2022). Our aim was not limited to fostering the application of AR in the cosmetics industry, but also to advance knowledge of human-machine interactions across different sectors.

1.2 Research model and hypotheses

To identify the factors influencing consumers' use of AR apps for purchasing cosmetics online, a theoretical framework is essential (Perea y Monsuwe et al., 2004; Riar et al., 2023). We adopted the Technology Acceptance Model (TAM), initially proposed by Davis in 1986, which aims to explain the determinants of new technology acceptance (Davis, 1986; Davis et al., 1989; Davis et al., 1992; Rese et al., 2017). TAM is considered one of the most widely used frameworks in examining marketing practices, also in AR applications (Du et al., 2022) as it helps explain its acceptance and potential future use (Rese et al., 2017). Several studies supported the idea that AR applications can influence consumer behavior by offering additional informational and interactive features, which may affect behavioral intentions (Childers et al., 2001; Pantano and Di Pietro, 2012; Dehghani et al., 2020; Romano et al., 2021). This study focused on the interaction between 4 TAM's dimensions, which are key factors that influence decisions regarding the adoption of technology: perceived informativeness (PI), perceived ease of use (PEOU), perceived enjoyment (PE), and behavioral intention (BI), In addition, we incorporated four additional constructs: makeup involvement (MI), perceived diagnosticity (PD), TRUST, and DOUBT.

The following subsections provide a description of each of the 8 variables, justifying their inclusion in the model and their roles as antecedents, mediators, and outcomes.

1.2.1 Perceived informativeness (PI)

Perceived Informativeness (PI) is a key factor in decision-making, providing consumers with relevant and useful product information that enhances clarity and helps achieve satisfactory product choices (Rese et al., 2017). PI positively impacts online purchase intention (Toraman, 2022; Kang et al., 2020), particularly in AR environments, where it emphasizes the functional and utilitarian aspects of the technology (Holdack et al., 2022). PI improves PEOU (Machdar, 2016) and perceived usefulness (Wixom and Todd, 2005). Additionally, higher PI positively influences PE, as the playful presentation of information increases hedonic value (Pantano and Di Pietro, 2012). In AR applications, PI and PE work together by providing useful digital information and enhancing the hedonic experience (Olsson et al., 2013). Recent studies show a direct positive effect of PI and PE on BI in the cosmetic domain (Oyman et al., 2022). Therefore, AR devices have the potential to provide more information, reduce uncertainty, and aid decision-making (Pantano and Di Pietro, 2012; Poushneh, 2018; Rese et al., 2017). As per TAM, PI is an important antecedent of AR usage.

1.2.2 Perceived ease of use (PEOU)

Perceived Ease of Use (PEOU) refers to the perceived ease with which individuals use technology, requiring minimal mental effort (Rouibah et al., 2011; Amin et al., 2014). It reduces learning costs and facilitates information seeking (Wang et al., 2014), promotes greater use of online platforms (Li and Yeh, 2010), and shapes positive attitudes toward technology adoption (Ahn et al., 2004). While PEOU is not always directly linked to BI (Kim and Kang, 2012; Chan and Teo, 2007), it remains a key factor in assessing user acceptance of online platforms (Amin et al., 2014). PEOU indirectly influences BI in online purchases (Wang et al., 2014) and predicts repurchase intentions (Rezaei and Amin, 2013). The mediating role of perceived enjoyment (PE) has been highlighted in the PEOU-BI relationship (Balog and Pribeanu, 2010; Holdack et al., 2022; Toraman, 2022), positioning PEOU as a crucial predictor in the AR framework of TAM (Holdack et al., 2022). However, high PEOU does not always predict system usage or improve attitudes, as shown in online banking (Al-Sharafi et al., 2016; Zheng and Li, 2020; Chang et al., 2015). In some cases, ease of use, especially in security-sensitive contexts, can signal liability or mistrust (Al-Sharafi et al., 2016), leading to doubts about the application (Sugandini et al., 2019). Thus, ease of use remains a critical factor in the engagement with AR tools and is considered an antecedent.

1.2.3 Behavioral intention (BI)

Behavioral Intention (BI) is a key factor for predicting the future use of a technology (Jackson et al., 1997). As noted by Oyman et al. (2022), it can be used to forecast AR application usage and it frequently emerges as an outcome in TAM models in marketing and online AR-based consumer experiences (e.g., Zhuang et al., 2020; Saleem et al., 2021; Park and Yoo, 2020) as it reflects the likelihood of engaging in a particular behavior (Ajzen and Fishbein, 1977; Turner et al., 2010). Oyman et al. (2022) found that both PI and PE positively and directly influence BI in the context of AR usage in the cosmetics domain. The positive relationship between PE and BI has been confirmed in previous studies (e.g., Qiu and Li, 2008), as well as the direct effect of PI on BI (e.g., Richard and Meuli, 2013). However, the role of PEOU in predicting BI is less consistent, with some studies suggesting direct (e.g., Lee et al., 2017), indirect (e.g., Revels et al., 2010; Wang et al., 2014) or negligible effects (e.g., Oyman et al., 2022; Hur et al., 2017; King and He, 2006). PE's role as a mediator between PEOU and BI has been discussed previously. Given its importance, BI is considered an outcome of the model, with PEOU and PI as potential predictors, and PE as a mediator.

1.2.4 Perceived enjoyment (PE)

Perceived Enjoyment (PE) refers to how enjoyable users find interacting with technology (Qiu and Li, 2008). Both PEOU and PI positively influence PE in virtual environments, contributing to positive attitudes toward technology and subsequent BI (Holdack et al., 2022). In AR or VR environments, PE plays a key role in technology acceptance (Pantano and Servidio, 2012; Balog and

Pribeanu, 2010). AR provides hedonic experiences like self-expression, entertainment, and stimulation (Olsson et al., 2013), which positively impact attitudes toward online purchases through virtual try-on technology (Zhang et al., 2019) and AR wearables (Holdack et al., 2022). Additionally, PE influences behavioral intention to use mobile media and shop online (Alalwan et al., 2018; Hasan et al., 2021; Dickinger et al., 2008). In this study, PE is considered a mediator.

1.2.5 Makeup involvement (MI)

Involvement refers to the subjective relevance, interest, and attentional resources directed toward a product category with personal significance and priority (Zaichkowsky, 1985; Zaichkowsky, 2010; Laurent and Kapferer, 1985). In the beauty and cosmetic sector, it is crucial for profiling customers (Stiehler and Jordaan, 2019; Guthrie et al., 2008) and refers to the level of attention and value consumers place on makeup use (Lee and Oh, 2018). Involvement is shaped by social, psychological, and functional values (Lichtenstein et al., 1988), enhanced by the self-expressive role of makeup (Bloch, 1982). As a result, Involvement drives consumers to engage more enthusiastically with products and to have higher expectations of both utilitarian and symbolic outcomes (Bloch and Richins, 1983). Scholars have emphasized different perspectives on involvement, including the role of rewards in interactions (Bloch and Bruce, 1984) and the motivation to fulfill functional and symbolic needs (Rothschild, 1984; Laurent and Kapferer, 1985). Involvement also has an emotional component, as highly involved consumers seek pleasure in product-related experiences (Venkatraman, 1989; Mathwick and Rigdon, 2004). Using TAM, Koufaris (2002) demonstrated the positive relationship between involvement and enjoyment in online shopping. Lee and Oh (2018) incorporated 'pleasure pursuit' into their model of *Makeup Involvement (MI)*, confirming its role as a potential outcome in cosmetics usage. Involvement helps protect products from negative evaluations (Dens and De Pelsmacker, 2010), increases willingness to pay (Steenkamp et al., 2010), enhances consumer brand identification (Stokburger-Sauer et al., 2012), and leads to product purchase (Olsen, 2007). Given its connection to reward and symbolic and utilitarian use, we expect MI to be a potential outcome of the model, directly linked to both PI and PEOU. Additionally, as MI is fostered by enjoyable experiences, we hypothesize that PE mediates the relationship between PEOU and PI.

1.2.6 Perceived diagnosticity (PD)

Perceived Diagnosticity (PD) refers to the effectiveness of a product experience in shaping opinions about a brand and its attributes (Kempf and Smith, 1998; Kempf, 1999). In virtual product experiences, PD reflects how effectively technology communicates useful, relevant, and helpful information for evaluating products (Jiang and Benbasat, 2004). A key factor in PD is the informativeness of the product or technology (Jiang and Benbasat, 2007; Chrimes et al., 2022; Yoo, 2020). In the AR context, Uhm et al. (2022) found higher PD levels compared to traditional web environments, due to greater information personalization. Yoo (2020) demonstrated that PD in AR is positively influenced by the accuracy, objectivity, and relevance of displayed information, aligning with the information quality of an information system (DeLone and McLean, 2003; Zheng et al., 2013). However, PEOU does not directly affect PD (Yoo, 2020), as usability can be influenced by personal characteristics (e.g., Aldás-Manzano et al., 2009) or task difficulties (e.g., Jiang and Benbasat, 2007), leading consumers to focus less on technical issues when evaluating diagnosticity (Park and Yoo, 2020). Nonetheless, Jiang and Benbasat (2004) found that PD increases when users perceive greater control over the environment, a factor also linked to PEOU (Venkatesh, 2000; Mathieson et al., 2001). Vividness (Chen et al., 2022) and visual quality (Yoo, 2020), along with interactivity with the product and environment (Chen et al., 2022), also enhance PD, contributing to enjoyment (Pantano et al., 2017). PD reduces perceived information asymmetry (Choe et al., 2008), increases perceived usefulness (Peukert et al., 2019), and strengthens the link between virtual brand experiences and purchase intentions (Gabisch and Gwebu, 2011), positively influencing behavioral intentions such as purchase and transaction intentions (Gabisch and Gwebu, 2011; Verhagen et al., 2016; Fang, 2012) and contributing to positive brand attitudes (Lee, 2012). In AR, PD is particularly important as it helps overcome the limited product interaction typical of online experiences (Yoo, 2020), and is driven by higher levels of tangibility (Verhagen et al., 2016). Based on these findings, we hypothesize that PD is an outcome of both PI and PEOU, with PE potentially serving as a mediator.

1.2.7 TRUST

TRUST is a key factor in consumer behavior and a crucial element in TAM models applied to AR (Khoshroo and Irani, 2024). It is defined as "a psychological state comprising the intention to accept vulnerability based on positive expectations of the intentions or behavior of another" (Rousseau et al., 1998, p. 395). TRUST helps reduce uncertainty, social complexity, and perceived risk by increasing the expectation of favorable outcomes (Pavlou, 2003; Bano and Siddiqui, 2022). In e-commerce, TRUST and PE are essential for online shopping quality (Ha and Stoel, 2009; Bilgihan, 2016; Kim and Peterson, 2017; Weisberg et al., 2011). Perceived usefulness and PEOU directly influence TRUST in online applications (Bano and Siddiqui, 2022; Beldad and Hegner, 2018; Ha et al., 2019; Hansen et al., 2018; Handoyo, 2024). TRUST also impacts BI in online banking (Rouibah et al., 2016) and mobile experiences (Hameed et al., 2024). AR technology in e-commerce enhances product interaction and facilitates shopping, with users' intention to use AR being shaped by performance expectations, PEOU, and hedonic motivation (Marto et al., 2023; Smink et al., 2019). TRUST plays a significant role in shaping performance expectations, reinforcing BI to use AR, and influencing the final perception of the tested product. Kim et al. (2010) found a positive relationship between the PI of a digital shopping experience and TRUST, confirming the former as a potential predictor. The authors also noted that entertainment during the experience positively shapes TRUST, reinforcing the role of PE in fostering positive attitudes (Chakraborty et al., 2003). Greater TRUST in AR for realistic product trials increases the likelihood of purchase, thus we hypothesized it as an outcome.

1.2.8 DOUBT

DOUBT is defined as "the subjective uncertainty people experience when evaluating the correctness of their decisions,

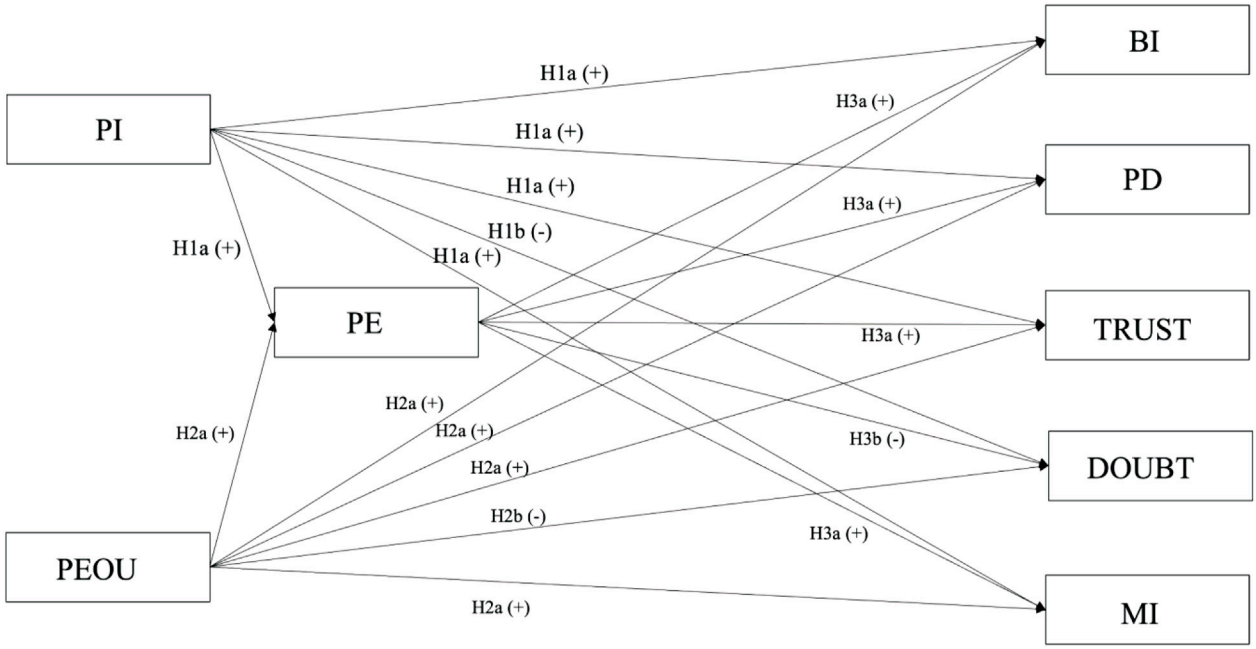


FIGURE 1
Research hypotheses. *Note*. PI, Perceived Informativeness; PEOU, Perceived Ease of Use; PE, Perceived Enjoyment; BI, Behavioural Intention; PD, Perceived Diagnosticity; TRUST, Trust; DOUBT, Doubt; MI = Make-up Involvement.

beliefs, or opinions" (Van de Calseyde et al., 2018, p. 98). It is also referred to as skepticism (Romani, 2006) and was included in the model due to its inverse relationship with TRUST, as highlighted by some scholars (e.g., Evans et al., 2021). Both DOUBT and TRUST play roles in decision-making processes (Evans et al., 2021), with the reverse relationship being particularly strong when exploring new products or using novel tools. In such cases, DOUBT significantly raises barriers to product usage (Naaz et al., 2021; Sääksjärvi and Morel, 2010). TRUST helps alleviate DOUBT and uncertainties in AR-mediated e-commerce (Manchanda and Deb, 2021), turning these into an incentive to adopt new technologies (Joerss et al., 2021). However, correlations have been found between DOUBT and PEOU (Rodrigues et al., 2024). In e-commerce, AR helps differentiate product options, reducing confusion, DOUBT, and anxiety (Barta et al., 2023b). As a result, AR can enhance the shopping experience by making it more engaging and increasing its PE (Basari and Dewanti, 2024). Thus, doubt was included as a potential outcome.

1.3 Research hypotheses

As summarised in Figure 1, we hypothesize that:

H1a - PI has a positive association with PE, BI, PD, TRUST, and MI.

H1b - PI has a negative association with DOUBT.

H2a - PEOU has a positive association with PE, BI, PD, TRUST, and MI.

H2b - PEOU has a negative association with DOUBT.

H3a - PE has a positive association with PE, BI, PD, TRUST, and MI.

H3b - PE has a negative association with DOUBT.

H4a - PE has a role as a mediator between PI as an antecedent and TRUST, MI, PD, DOUBT, and BI as outcomes.

H4b - PE has a role as a mediator between PEOU as an antecedent and TRUST, MI, PD, DOUBT, and BI as outcomes.

2 Materials and methods

2.1 Participants and procedure

We conducted a self-report questionnaire using an online platform (Google Forms). The sample consisted of 634 Italian cosmetics consumers, primarily female (82%), with an average age of 30 years (SD = 12.72, range: 18–75). Most of them were unmarried (67%), held a university degree (42%), and were either students (44%) or employees (25%).

Before completing the questionnaire, participants provided informed consent that acknowledged their voluntary participation and ensured their anonymity. The study adhered to the principles outlined in the Declaration of Helsinki (World Medical Association, 2013) and complied with the General Data Protection Regulation. Ethical approval was obtained from the IULM Ethical Committee (Protocol 0073876 of 19/09/24). The questionnaire was administered from February 2024 to April 2024.

2.2 Measures

PI, PEOU, PE, BI, PD, TRUST, DOUB and MI dimensions were averaged from multiple items, with participants indicating their level